

Handling Missing Data in ETL

Detailed Assignment Solution

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Platform: PwSkills

This document contains detailed theoretical explanations and practical solutions related to handling missing data in ETL pipelines.

Question 1: What are the most common reasons for missing data in ETL pipelines?

Missing data is a common issue in ETL pipelines. Several factors can cause data to be missing during extraction, transformation, or loading.

1. Data Entry Errors – Users sometimes skip fields while filling forms.
2. System Failures – Technical failures in applications or sensors can prevent data from being recorded.
3. Data Integration Issues – When combining datasets from multiple sources, some attributes may not exist in all sources.
4. Data Corruption – During file transfer or database migration some values may get corrupted or lost.
5. Privacy Concerns – Users may intentionally avoid sharing sensitive information such as income or age.

Understanding the reason for missing data is important because the handling technique should depend on the root cause.

Question 2: Why is blindly deleting rows with missing values considered a bad practice in ETL?

Blindly deleting rows with missing values can negatively impact data quality and analysis.

Loss of Valuable Information – Even if one column has missing values, other columns may contain useful data.

Reduced Dataset Size – Removing too many rows decreases the dataset size which can weaken statistical analysis.

Bias in Data – If certain groups have more missing values, deleting them can introduce bias.

Incorrect Insights – Business decisions made on incomplete or biased datasets can be misleading.

Therefore, analysts should carefully analyze missing patterns and choose appropriate techniques such as imputation, forward fill, or flagging.

Question 3: Explain the difference between Listwise Deletion and Column Deletion.

Feature	Listwise Deletion	Column Deletion
Definition	Rows with missing values are removed	Entire column removed
Data Loss	Only some rows lost	Complete variable lost
Usage	When few rows missing	When column mostly missing
Impact	Dataset smaller	Feature removed from analysis

Listwise deletion is appropriate when only a small number of rows contain missing values. Column deletion is suitable when a column contains a very large percentage of missing values and is not useful for analysis.

Question 4: Why is median imputation preferred over mean imputation for skewed data such as income?

Income data is usually skewed because a few individuals earn extremely high salaries compared to the majority.

Mean imputation calculates the average value. However, extreme values (outliers) can significantly distort the mean.

Median represents the middle value of a sorted dataset and is not affected by extreme outliers.

For example: If incomes are 30000, 35000, 40000, and 1000000, the mean becomes unrealistically high, while the median remains closer to the typical income.

Therefore, median imputation is more reliable for skewed datasets.

Question 5: What is forward fill and in what type of dataset is it most useful?

Forward fill is a technique where missing values are replaced with the previous valid observation in the dataset.

Example:

Day	Sales
Day1	100
Day2	NaN
Day3	120

After forward fill, the missing value for Day2 becomes 100. This method is most useful in time-series datasets such as stock prices, weather measurements, sensor readings, and daily sales data.

Question 6: Why should flagging missing values be done before imputation in an ETL workflow?

Flagging missing values means creating a separate column that indicates whether a value was originally missing.

Reasons:

1. Preserves original information about missing data.
2. Helps machine learning models detect patterns related to missing values.
3. Allows analysts to understand the extent of missing data.
4. Supports better business analysis.

Question 7: How can missing income data provide business insights?

Missing income values can reveal useful behavioral patterns.

Customers may avoid sharing income due to privacy concerns.

It may indicate specific demographic groups who hesitate to provide financial information.

Businesses can redesign data collection forms or provide incentives to encourage customers to share information.

Thus missing data can itself provide meaningful insights.

Question 8: Listwise Deletion – Remove rows where Region is missing.

Customer_ID	Name	City	Monthly_Sales	Income	Region
101	Rahul Mehta	Mumbai	12000	65000	West
104	Neha Singh	Delhi	NaN	NaN	North
102	Anjali Rao	Bengaluru	NaN	NaN	South
105	Amit Verma	Pune	18000	58000	NaN
107	Pooja Das	Kolkata	14000	NaN	East
103	Suresh Iyer	Chennai	15000	72000	South
106	Karan Shah	Ahmedabad	NaN	61000	West
108	Riya Kapoor	Jaipur	16000	69000	North

Row with missing region: Customer ID 105 (Amit Verma). After listwise deletion total rows become 7 instead of 8.

Question 9: Apply Forward Fill on Monthly_Sales.

Customer	Monthly Sales Before
Rahul	12000
Neha	NaN
Anjali	NaN
Amit	18000
Pooja	14000
Suresh	15000
Karan	NaN
Riya	16000

Customer	Monthly Sales After Forward Fill
Rahul	12000
Neha	12000
Anjali	12000
Amit	18000
Pooja	14000
Suresh	15000
Karan	15000
Riya	16000

Question 10: Create Income Missing Flag.

Customer	Income	Income_Missing_Flag
Rahul Mehta	65000	0
Neha Singh	NaN	1
Anjali Rao	NaN	1
Amit Verma	58000	0
Pooja Das	NaN	1
Suresh Iyer	72000	0
Karan Shah	61000	0
Riya Kapoor	69000	0

Total customers with missing income: 3